

Grade 6 Mathematics Core Concept Mastery & Problem Solving Exam (Set 10)

Name:	Class:	Student ID:	Time: 50-60 Mins	Total: 100 Pts	Score:
_____	_____	_____			_____

Part 1: Conceptual Precision & Mathematical Truth Statements

5 Questions, Total 20 Pts

1. Place Value Fundamentals (4 pts)

- (1) To determine whether an integer is divisible by 8 or 125, we only need to inspect its last _____ digits, because $1000 = \rule{1.5cm}{0.4pt}$.
- (2) List all 3-digit positive multiples of 125 below 1000 (list all):
- _____

2. Counterexamples & Logical Rigor (4 pts)

- (1) "All multiples of 5 are composite numbers." (Status: _____, Counterexample: _____)
- (2) "A composite number is always strictly greater than a prime number." (Status: _____, Counterexample: _____)

3. Pure Cycles vs Range Definition (4 pts)

- (1) Compare values: $1.\dot{2}\dot{3}$ _____ $1.\dot{2}\dot{3}$. (Fill in ">", "<", or "=")
- (2) In a class exam, the highest score was 98 and the lowest was 62. The score Range is _____. If 18 students scored 85 (the highest frequency), the Mode is _____.

4. The Pizza & Cake Scaling Models (4 pts)

- (1) A "12-inch" pizza refers to its diameter. The area of a 12-inch pizza is _____ times that of a 6-inch pizza. Hence two 6-inch pizzas provide only _____ (fraction) of the original pizza!
- (2) A bakery sells an 8-inch cube cake. If replaced with 4-inch cube cakes, it takes exactly _____ smaller cakes to match the volume of one 8-inch cake.

5. Periodicity Deduction (4 pts)

- (1) $2^4 = \rule{1.5cm}{0.4pt}$.
- (2) The unit digit of 2^n cycles every 4 steps: 2, 4, 8, 6. Since $2026 \div 4 = 506$ remainder _____, the unit digit of 2^{2026} is _____; similarly, the unit digit of 3^{2026} is _____.

Part 2: Dimensional Transformations & Metric Conversions

4 Questions, Total 20 Pts

6. 1D, 2D, and 3D Metric Scaling (5 pts)

- (1) $7.04\text{ m}^2 = \rule{1.5cm}{0.4pt}\text{ dm}^2$ (2) $240\text{ cm}^2 = \rule{1.5cm}{0.4pt}\text{ dm}^2$
- (3) $1\text{ m}^3 = \rule{1.5cm}{0.4pt}\text{ dm}^3 = \rule{1.5cm}{0.4pt}\text{ cm}^3$ (4) $6\text{ tonnes } 60\text{ kg} = \rule{1.5cm}{0.4pt}\text{ tonnes}$

7. Spatial Physical Interpretation (5 pts)

- A cuboid has a volume of 24 cm^3 and a vertical height of 3 cm.
- (1) Calculate: $24\text{ cm}^3 \div 3\text{ cm} = \rule{1.5cm}{0.4pt}$.
- (2) In terms of geometry and physical dimensions, explain why division causes "dimensionality reduction", and identify which faces of the cuboid this area represents (and why it cannot be the front or side faces):

Explanation:

8. Fast Mental Verification (5 pts)

(1) Mental multiplication: $240 \times 3.5 = \underline{\hspace{2cm}}$.

(2) Use powers of 10 to evaluate $2,400,000 \times 0.000035$ cleanly without manual column counting:

(3) A student mistakenly calculated $1.5 \times 36 = 5.4$. Explain how to verify that this result is incorrect using estimation:

Estimation Analysis:

9. Trapezoid Area Derivation (5 pts)

A diagonal divides a trapezoid with top base a , bottom base b , and height h into two triangles.

(1) Area of lower triangle $S_1 = \underline{\hspace{2cm}}$; Area of upper triangle $S_2 = \underline{\hspace{2cm}}$.

(2) Use factoring out the common factor $\frac{1}{2}h$ (reverse distributive property) to prove $S = \frac{1}{2}(a+b)h$:

Part 3: Rigorous Calculation, Number Theory & Telescoping

4 Questions, Total 20 Pts

10. Careful Fraction Addition (5 pts)

(1) Calculate: $2\frac{1}{3} + 3\frac{5}{6}$

(2) Calculate: $\frac{1}{6} + \frac{1}{8}$

11. Prime Power Extremum Rule (5 pts)

Two positive integers are given in prime factorization form: $E = 2^{10} \times 3^4$, $F = 2^6 \times 3 \times 5^5$.

(1) Find $\text{HCF}(E, F)$ in prime factor index form: $\underline{\hspace{2cm}}$.

(2) Find $\text{LCM}(E, F)$ in prime factor index form: $\underline{\hspace{2cm}}$.

12. Factor Redundancy (5 pts)

Given positive integers 12 and 28:

(1) Use short division to find $\text{GCD}(12, 28) = \underline{\hspace{1cm}}$ and $\text{LCM}(12, 28) = \underline{\hspace{1cm}}$.

(2) Calculate: $12 \times 28 = \underline{\hspace{2cm}}$; $\text{GCD}(12, 28) \times \text{LCM}(12, 28) = \underline{\hspace{2cm}}$.

(3) Briefly explain why the product of any two positive integers always equals the product of their GCD and LCM:

Mathematical Rationale:

13. Fractional Difference Expansion (5 pts)

(1) Prove the identity: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ by expanding the right-hand side:

(2) Evaluate the classroom summation using telescoping cancellation:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{2025 \times 2026}$$

Final Value =

Part 4: 3D Geometry & Combined Solid Surface Area

2 Multi-part Problems, Total 20 Pts

14. Double-overlap Deduction (10 pts)

A large wooden cuboid measures $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$. A small wooden cube with edge length 2 cm is glued onto the center of one of its faces to form a composite solid. All exposed outer surfaces are to be spray-painted.

(1) Before gluing, the independent surface area of the cuboid is _____ cm^2 ; the cube is _____ cm^2 . (2 pts)

(2) When joined, the number of hidden overlapping faces is _____, with total hidden overlap area of _____ cm^2 . (2 pts)

(3) Calculate the total exposed surface area of the composite solid: (3 pts)

Calculation:

Ans: Exposed Surface Area = _____ cm^2 .

(4) If 1 cm^2 of surface requires 200 mg of paint, how many grams (g) of paint are needed in total? (3 pts)

Calculation:

Ans: Total paint needed = _____ grams.

15. (10 pts)

Two identical wooden cuboids measuring $5 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$ are joined together face-to-face in three different configurations.

(1) Independent surface area of one cuboid is _____ cm^2 ; sum of two cuboids before gluing is _____ cm^2 . (2 pts)

(2) **Configuration A (glue 3×5 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(3) **Configuration B (glue 2×5 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(4) **Configuration C (glue 2×3 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(5) A student calculated that both Configuration B and Configuration C yield a surface area of 104 cm^2 .

Determine whether this conclusion is correct through rigorous calculation, provide your reasoning, and state the general inverse law between overlap area and resulting surface area: (2 pts)

Proof & Inverse Law:

Part 5: Integrated Word Problems & Modulo Divisibility Finale

3 High-Order Problems, Total 20 Pts

16. (6 pts)

A runner completes a 1000 m distance on a circular track at a constant speed in 20 minutes.

(1) What is the runner's speed in meters per minute (m/min), and in kilometers per hour (km/h)? (3 pts)

Calculation:

Ans: Speed = _____ m/min = _____ km/h .

(2) What fraction of the total journey did the runner cover during the final 7 minutes? (3 pts)

Calculation:

Ans: Fraction of journey = _____.

17. Magnitude Ordering (6 pts)

Four distinct positive numbers A, B, C, D satisfy:

$$A - 2020 = B + 2021 = C - 2022 = D + 2023 = k \quad (k > 0)$$

(1) Explain algebraically or logically why C is the greatest and D is the least: (3 pts)

Intuitive Rationale:

(2) Express A, B, C, D algebraically in terms of k , and rank them from greatest to least using ">": (3 pts)

Expressions: $A =$ _____, $B =$ _____, $C =$ _____, $D =$ _____

Ordering: _____

18. 6-Digit Number Challenge (8 pts)

A 6-digit number $\overline{91X93Y}$ (X, Y are single digits, $0 \leq X, Y \leq 9$) is divisible by 65.

(1) Since $65 = 5 \times 13$, the number must be divisible by both 5 and 13. The possible values for unit digit Y are _____. (1 pt)

(2) Knowing that $1001 = 7 \times 11 \times 13$ is a multiple of 13, prove that $1000X \equiv -X \pmod{13}$: (2 pts)

(3) Decompose $\overline{91X93Y} = 910000 + 1000X + 930 + Y$. Since $910000 = 13 \times 70000 \equiv 0 \pmod{13}$ and $930 = 71 \times 13 + 7 \equiv 7 \pmod{13}$, state the necessary and sufficient divisibility condition modulo 13: (_____) is divisible by 13. (2 pts)

(4) Solve for the unknown digits through branch analysis: (3 pts)

[Teacher Answer Key & Scoring Guide] (Teacher Reference)

Part 1: Conceptual Precision (20 pts)

- (1) 3; 8×125 (1 pt each); (2) 125, 250, 375, 500, 625, 750, 875 (2 pts for all 7).
- (1) False, counterexample: 5 is prime (2 pts); (2) False, counterexample: 4 is composite and $4 < 5$ (or $9 < 11$) (2 pts).
- (1) $>$ ($1.2333... > 1.2323...$, 2 pts); (2) Range = 36, Mode = 85 (1 pt each).
- (1) 4 times; $1/2$ (1 pt each); (2) 8 (2 pts, volume ratio is $(8/4)^3 = 8$).
- (1) 16 (2 pts); (2) Remainder 2, unit digit of 2^{2026} is 4, unit digit of 3^{2026} is 9 (2 pts).

Part 2: Dimensional Transformations & Units (20 pts)

- (1) 704; (2) 2.4; (3) 1000, 1,000,000; (4) 6.06 (1 pt each).
- (1) 8 cm^2 (1 pt); (2) Volume is $3D (L \times W \times H)$. Dividing by height H removes the vertical z-axis dimension, leaving the 2D horizontal plane ($L \times W$), which corresponds precisely to the top and bottom faces. The front and side faces both contain height H , so they cannot be represented once height is divided out (4 pts). Dividing by length 4cm yields 6 cm^2 (left/right side faces).
- (1) 840 (1 pt); (2) $240 \times 10^4 \times 3.5 \times 10^{-5} = 840 \times 10^{-1} = 84$ (2 pts); (3) Estimation: $1.5 \approx 2$, $36 \approx 30 \implies 2 \times 30 = 60$. The value 5.4 differs by an entire factor of 10, immediately exposing a misplaced decimal point (2 pts).
- (1) $\frac{1}{2}bh$, $\frac{1}{2}ah$ (1 pt each); (2) $S = \frac{1}{2}ah + \frac{1}{2}bh = \frac{1}{2}(a+b)h$ via factoring out $\frac{1}{2}h$ (3 pts).

Part 3: Rigorous Calculation & Number Theory (20 pts)

- (1) $2\frac{2}{6} + 3\frac{5}{6} = 5\frac{7}{6} = 6\frac{1}{6}$ or $\frac{37}{6}$ (3 pts); (2) $\frac{4}{24} + \frac{3}{24} = \frac{7}{24}$ (2 pts).
- (1) $\text{HCF} = 2^6 \times 3$ (2 pts); (2) $\text{LCM} = 2^{10} \times 3^4 \times 5^5$ (3 pts).
- (1) $\text{GCD} = 4$, $\text{LCM} = 84$ (1 pt each); (2) $12 \times 28 = 336$, $4 \times 84 = 336$ (1 pt each); (3) In direct multiplication, shared prime factors (GCD) are counted twice. Dividing by one copy of GCD yields the LCM, hence $a \times b = \text{GCD} \times \text{LCM}$ (1 pt).
- (1) $\frac{1}{n} - \frac{1}{n+1} = \frac{(n+1)-n}{n(n+1)} = \frac{1}{n(n+1)}$ (2 pts); (2) $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots + (\frac{1}{2025} - \frac{1}{2026}) = 1 - \frac{1}{2026} = \frac{2025}{2026}$ (3 pts).

Part 4: 3D Geometry & Surface Area (20 pts)

- (1) 94, 24 (1 pt each); (2) 2 hidden faces, overlap area = $2 \times 4 = 8 \text{ cm}^2$ (1 pt each); (3) $94 + 24 - 8 = 110 \text{ cm}^2$ (3 pts); (4) $110 \times 200 \text{ mg} = 22000 \text{ mg} = 22 \text{ g}$ (3 pts).
- (1) 62, 124 (1 pt each); (2) 30, 94 (1 pt each); (3) 20, 104 (1 pt each); (4) 12, 112 (1 pt each); (5) Config B overlap is $2 \times 10 = 20 \implies 104 \text{ cm}^2$; Config C overlap is $2 \times 6 = 12 \implies 112 \text{ cm}^2 \neq 104$. Law: smaller overlap area leaves larger exposed surface area (2 pts).

Part 5: Integrated Word Problems & Modulo Divisibility (20 pts)

- (1) Speed = $1000/20 = 50 \text{ m/min}$; $50 \times 60/1000 = 3 \text{ km/h}$ (3 pts); (2) $7/20$ (3 pts).
- (1) C spent the most money and still equals the others $\implies C$ started with the most; D had to receive the most to catch up $\implies D$ started with the least (3 pts); (2) $A = k + 2020$, $B = k - 2021$, $C = k + 2022$, $D = k - 2023 \implies C > A > B > D$ (3 pts).
- (1) $Y \in \{0, 5\}$ (1 pt); (2) $1000X = 1001X - X = 13 \times 77X - X \equiv -X \pmod{13}$ (2 pts); (3) Remainder 7; Divisibility requires $7 + Y - X \equiv 0 \pmod{13}$ (2 pts); (4) Branch 1 ($Y = 0$): $7 - X \equiv 0 \implies X = 7$. Branch 2 ($Y = 5$): $12 - X \equiv 0 \implies X = 12$ (invalid) or $12 - X = 13 \implies X = -1$ (invalid). Unique solution: $X = 7$, $Y = 0$, number is 917930 (3 pts).